DEEP LEARNING FINAL PROJECT

Finding the Right Law, in Khmer

Can a model match everyday Khmer questions with the correct legal articles?

EXAMPLE USER QUESTION

តើកិច្ចសន្យាបង្កើតឡើងដោយរបៀបណា?

1,976	4	200
LEGAL ARTICLES	NEURAL APPROACHES	NATURAL TEST QUESTIONS

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A keyword match is not always a meaning match

The question and the law can describe the same idea using different words.

តើកិច្ចសន្យាបង្កើតឡើងដោយរបៀបណា?

BM25 / KEYWORD SEARCH

Finds overlapping words

One recorded top match was Article 623 — about dividing proceeds, not forming a contract.

NEURAL SEMANTIC RETRIEVAL

Compares meaning

A trained encoder maps questions and articles into vectors and ranks them by similarity.

Goal: retrieve the relevant article, not generate a new law.

First, we had to make the Khmer PDFs readable

Open Development Cambodia | Official Khmer codes | CC-BY-SA-4.0 | Limon font conversion

RAW EXTRACTED TEXT

maRta 336>- karbegáltkic©snña...

AFTER LIMON → UNICODE CONVERSION

មាត្រា ៣៣៦.- ការបង្កើតកិច្ចសន្យា...

Official PDFs
Source documents

→

Convert Limon
Legacy font mapping

→

Normalize Khmer
Character ordering

→

Split into articles
Preserve metadata

1,304 Civil Code articles

672 Criminal Code articles

We had legal articles, but not enough question–answer pairs

Article titles provide weak supervision; 100 Civil and 100 Criminal questions form the main test set.

WEAK SUPERVISION: A TRAINING PAIR

Article title → Article body

We removed titles from article bodies to avoid trivial text matching.

1,176	147	148
TRAIN	VALIDATION	TITLE TEST

Separate main benchmark: 200 natural Khmer questions · 261 reference articles excluded from training.

Four ways to teach a model to retrieve legal articles

Each approach produces embeddings for queries and documents.

A1 · START FROM SCRATCH

BiLSTM

Khmer word tokens

↓

1-layer BiLSTM

↓

512-d vector

1.97M trainable

A2 · FREEZE THE BACKBONE

XLM-R probe

Pretrained XLM-R

↓

Frozen backbone

↓

Linear head

0.59M trainable

A3 · ADAPT ALL LAYERS

XLM-R fine-tune

Pretrained XLM-R

↓

12 transformer layers

↓

768-d vector

278.04M trainable

A4 · KHMER PRETRAINING

PrahokBART

Khmer pretrained BART

↓

Encoder + pooling

↓

512-d vector

35.83M trainable

Contrastive learning: bring matching pairs closer

The loss rewards the correct question–article match over other articles in the batch.



One correct pair per row

Other articles in the batch act as negative examples.

A3 batch size: 16 · 15 in-batch negatives

Similarity → InfoNCE loss (τ = 0.05) → Backpropagation → Updated model weights

Experimental setup and compute

Fixed seed 42; shared splits and evaluation; model selection uses validation MRR@10.

Approach	Trainable params	Training time	Hardware
A1 BiLSTM	1.97M	2035.3 s	CPU
A2 XLM-R probe	0.59M	5.9 s	CPU
A3 XLM-R fine-tune	278.04M	1129.3 s	GPU (Tesla T4)
A4 PrahokBART	35.83M	1602.2 s	CPU (14t, fp32)

A3: AdamW; LR {1e-5, 2e-5, 3e-5} x weight decay {0, 0.01}; 5 epochs.

Batch 16; temperature 0.05; max length 256; warmup then cosine learning-rate schedule.

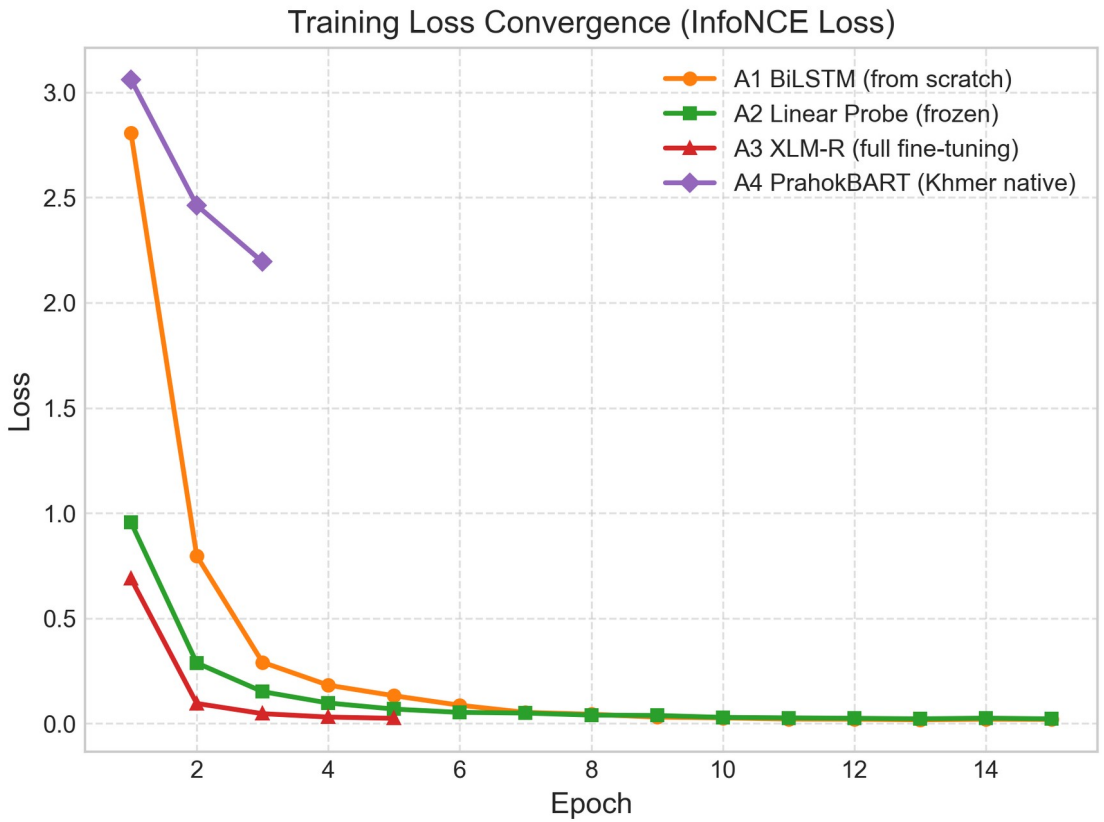
A2 time covers the head training on cached embeddings; excludes backbone encoding.

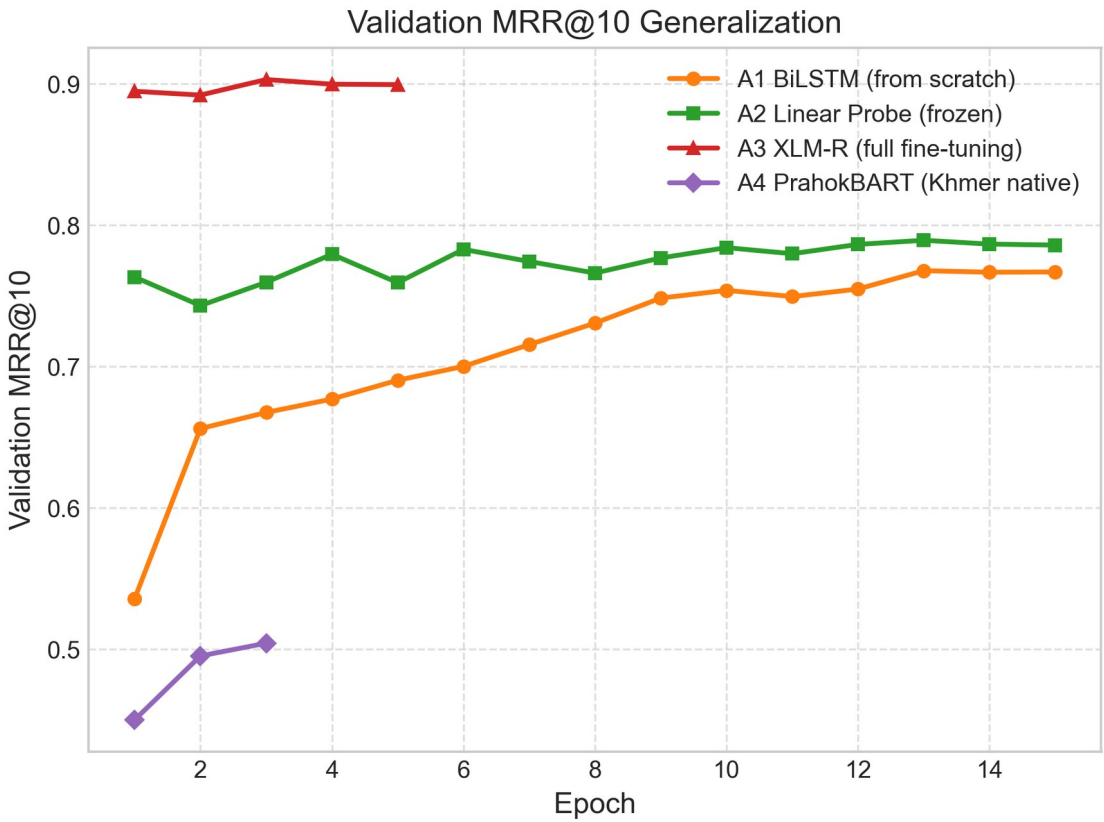
Times use different hardware and are not a controlled speed comparison.

Sources: results/metrics/summary.csv and results/tuning/a3.csv.

The learning curves told us which approaches were working

We tuned hyperparameters using validation data, not the natural-question test set.



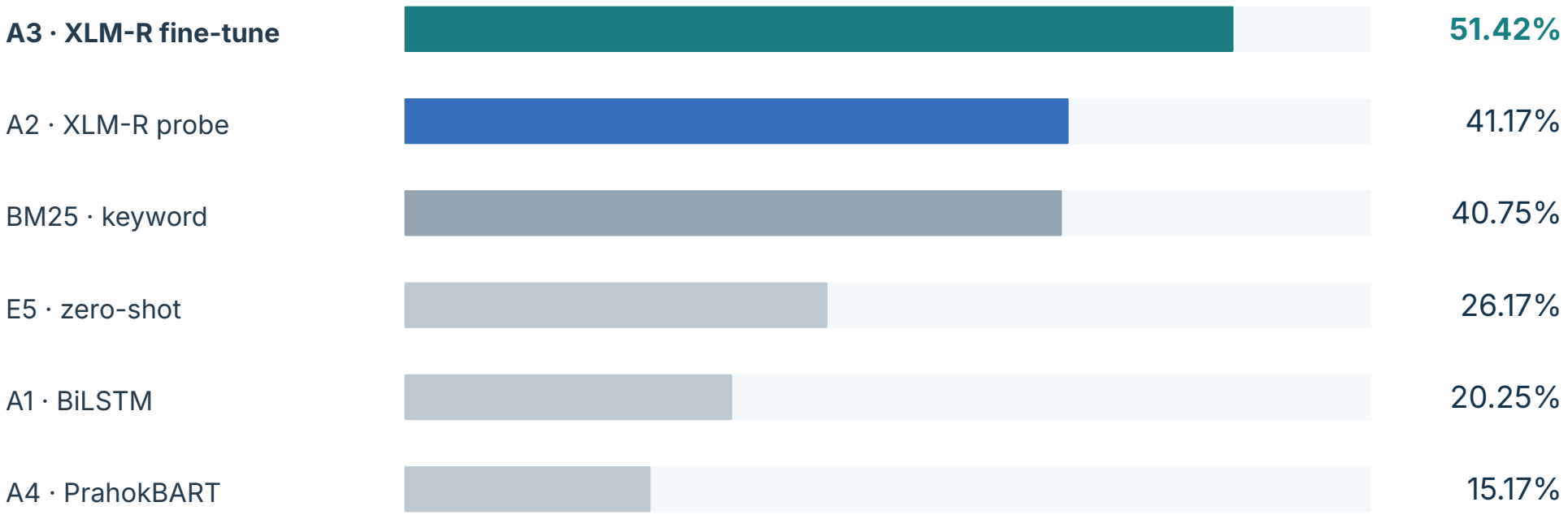


A3: 6 LR/weight-decay runs · selected LR 2×10^{-5}, WD 0.01 · validation MRR@10 0.9095 · Tesla T4

Full fine-tuning performed best on natural Khmer questions

All systems were evaluated against the same 200-question benchmark.

RECALL@5 — SHARE OF RELEVANT ARTICLES IN TOP FIVE



A3 VS BM25

+10.67

pp Recall@5

+44.1%

relative MRR

MRR@10: A3 0.4827 · BM25 0.3350 | Recall@5 difference: 10.67 percentage points

EXPERIMENTAL EVIDENCE

Results on the same 200 Khmer questions

All 1,976 articles are candidates. No statute filter is applied in the benchmark.

System	Recall@1	Recall@5	MRR@10	Hit@5
BM25	0.2025	0.4075	0.3350	0.4700
E5 zero-shot	0.1625	0.2617	0.2588	0.3200
A1 BiLSTM	0.0875	0.2025	0.1724	0.2450
A2 XLM-R probe	0.2400	0.4117	0.3591	0.4750
A3 XLM-R fine-tune	0.3375	0.5142	0.4827	0.5700
A4 PrahokBART	0.0525	0.1517	0.1035	0.1800

Recall@5: mean fraction of relevant articles recovered; Hit@5: any relevant article found.

MRR@10: reciprocal rank of the first relevant article, averaged over questions.

Source: results/metrics/summary.csv. BM25 and zero-shot E5 are additional baselines.

The same model got 99% on titles, but 51% on real questions

A strong score on one type of query did not transfer to another type.

TITLE-STYLE QUERIES

148 held-out titles

99.32%

Recall@5

NATURAL KHMER QUESTIONS

200 benchmark questions

51.42%

Recall@5

We trained mostly on legal titles, but people ask questions in very different ways.

Even the best model still makes important mistakes

Heuristic categories for 86 A3 misses; vocabulary mismatch is a residual label, not a proven cause.

60.5%

Vocabulary mismatch

The question uses different wording from the article.

19.8%

Near-misses

The right article appeared at ranks 6–10.

17.4%

Multiple articles

A question may need more than one provision.

Limits: two codes, 200 questions, one seed; manual extraction error rate still unmeasured.

Can we retrieve the right article from 1,976 candidates?

A controlled example from the held-out natural-question benchmark.

BENCHMARK QUESTION · TQ-CIV-009

តើនរណាខ្លះអាចស្នើសុំឱ្យតុលាការប្រកាសបើកអាណាព្យាបាលទូទៅ?

REFERENCE ANSWER

Civil Code · Article 24

BM25	A3 · Fine-tuned XLM-R
Keyword search	Semantic search

Live check: same query · top 5 · all statutes · compare Article 24 ranking.

What I learned from comparing four neural retrievers

Three takeaways, and one clear next step.

01 **Pretraining helped**

The from-scratch BiLSTM struggled on a small dataset.

02 **Fine-tuning won**

A3 was the strongest model on natural questions.

03 **Data still mattered**

99.32% title Recall@5 did not mean 99% on real questions.

Next: train with more natural Khmer questions and harder negative examples.

Model specifications

Extra details if the lecturer asks about architecture or parameters.

	A1 BiLSTM	A2 Probe	A3 Fine-tune	A4 BART
Pretraining	None	E5-base	E5-base	Khmer BART
Trainable params	1.97M	0.59M	278.04M	35.83M
Embedding size	512	768	768	512
Learning method	From scratch	Frozen backbone	All layers	Encoder fine-tune
Tokenization	Khmer words	SentencePiece	SentencePiece	Khmer subwords

How A3 was tuned

Six learning-rate / weight-decay configurations, selected by validation MRR@10.

VALIDATION MRR@10

	Weight decay 0.00	Weight decay 0.01
LR 1×10^{-5}	0.9023	0.9023
LR 2×10^{-5}	**0.9095**	**0.9095**
LR 3×10^{-5}	0.9082	0.9081

Selected: LR 2e-5, WD 0.01; tied at reported precision with WD 0.00. Best epoch 4 of 5.

Statistical comparison

Paired non-parametric bootstrap on the 200 natural benchmark questions.

Comparison	Metric	Difference	95% CI	p-value
A3 vs BM25	Recall@5	**+0.1067**	[+0.0500, +0.1633]	0.0002
A3 vs BM25	MRR@10	**+0.1477**	[+0.0962, +0.1993]	0.0001
A2 vs BM25	Recall@5	**+0.0042**	[-0.0567, +0.0625]	0.9031
A3 vs A2	Recall@5	**+0.1025**	[+0.0492, +0.1575]	0.0001

These results support an improvement on this benchmark; they do not guarantee correct legal retrieval for every question.

Reproducing and inspecting the system

Files to open if the lecturer asks about training, leakage, or the demo.

Data splits	src/dl/prepare_splits.py
Neural models	src/dl/models/
Contrastive objective	src/dl/losses.py
Training & tuning	src/dl/experiments/tune_a3.py
Evaluation & metrics	src/dl/evaluate.py · src/dl/metrics.py
Live Streamlit retriever	src/interfaces/ui/app.py

Demo prerequisite: A3 checkpoint and matching corpus embeddings loaded; never present BM25 fallback as A3.